PROJECT PLANNING PHASE

Electric Vehicle Market Performance & Insights

Project plan, work breakdown, and data preprocessing record

1. Planning Objective

This phase organizes the work required to deliver the EV market analytics project in Tableau Public. It records the main tasks, dependencies, tools, expected outputs, and data-preprocessing work completed before dashboard design and development. The plan is intended to keep the work structured and avoid repeating the detailed problem statement and requirements documented in earlier phases.

2. Work Breakdown and Deliverables

Work Package	Main Activities	Deliverable / Output
1. Data preparation	Inspect source files, check structure and quality, standardize text fields, identify duplicate and missing values.	Preprocessed combined CSV, quality report, preprocessing notes
2. Requirement confirmation	Confirm useful fields, metric definitions, filters, and analysis questions.	Agreed scope and metric list
3. Dashboard design	Plan KPI area, chart placement, filters, labels, and navigation.	Dashboard layout and chart plan
4. Tableau development	Connect data, build worksheets, assemble dashboard, configure interactions.	Interactive Tableau dashboard
5. Story and insights	Organize important views into a clear narrative and summarize observations.	Tableau Story and key findings
6. Validation and testing	Check totals, calculations, filters, labels, readability, and responsiveness.	Test results and corrections
7. Publication and handover	Publish/share the dashboard and prepare project documentation and demonstration flow.	Published project and supporting documents

3. Data Preprocessing Completed

The supplied train and test CSV files were inspected and combined for exploratory visualization. The following records summarize the preprocessing already carried out.

Item	Before / Source	Result / Action
Training dataset	ev_market_train.csv — 200,000 rows, 37 columns	Inspected as one source file
Testing dataset	ev_market_test.csv — 100,000 rows, 37 columns	Inspected for structure compatibility
Combination	300,000 total source rows	Combined into ev_market_preprocessed_combined.csv
Duplicate handling	One exact duplicate identified	Removed one duplicate; combined result contains 299,999 rows
Source tracking	Two input files combined	Added record_source field, giving 38

		columns in the output
Text cleanup	Text fields may contain extra whitespace or blank strings	Trimmed surrounding whitespace and converted blank text values to missing
Missing values	Missing values present in source data	28,507 missing cells remain; genuine missing numeric values were preserved rather than automatically imputed

4. Preprocessing Files and Their Purpose

- **ev_market_preprocessed_combined.csv —** Combined dataset with 299,999 rows and 38 columns, prepared for exploratory visualization.
- **ev_market_data_quality_report.csv —** Quality summary covering missing values, data types, and distinct-value counts.
- **ev_market_preprocessing_notes.txt —** Notes describing preprocessing decisions and the resulting dataset.

Important usage note: The combined train/test dataset is intended for exploratory visualization only where the project rules permit it. If the original train/test split has a modelling or evaluation purpose, keep the files separate for that purpose. Before final publication, review the remaining missing values and confirm that every KPI uses the correct field, unit, aggregation, and denominator.

5. Proposed Schedule and Dependencies

The sequence below is a relative plan rather than a fixed calendar schedule. Actual dates can be added to the project tracker once the submission deadline is confirmed.

Order	Phase / Task	Dependency or Completion Check
1	Data review and preprocessing	Source files inspected; output and quality notes saved
2	Requirement confirmation	Available fields mapped to intended questions and metrics
3	Project design	Charts, dashboard layout, and filters selected
4	Dashboard development	Worksheets and dashboard assembled in Tableau
5	Story and findings	Views arranged into a clear sequence
6	Testing and refinement	Calculations, filters, labels, and usability checked
7	Publication and documentation	Final dashboard link and project files ready

6. Tools and Resources

- Tableau Public — data visualization, dashboard building, Story creation, and publication.
- CSV source files — the supplied EV market train and test datasets.
- Preprocessing outputs — combined CSV, data-quality report, and preprocessing notes.
- Documentation tools — Word for phase documents and GitHub or another suitable repository for project files.

7. Risks and Mitigation

Potential Risk	Planned Mitigation
Missing or inconsistent values	Review the quality report; define field-specific handling

	and disclose important limitations.
Train/test files have different meanings or structures	Check schemas and project rules before combining; retain source tracking and preserve original files.
Misleading KPI calculations	Document each metric definition, units, aggregation, and filtering behaviour; validate totals.
Slow dashboard interactions	Limit unnecessary marks and complex calculations; test filters and dashboard responsiveness.
Unavailable geographic or time fields	Use maps or trend charts only if valid fields are present; otherwise choose an appropriate alternative.

8. Completion Criteria for the Planning Phase

- Major work packages and outputs have been identified.
- Completed preprocessing steps and output files are documented.
- Dependencies and key data risks have been recorded.
- The next step—Project Design—can proceed with a clear understanding of the prepared data and planned deliverables.